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ORIGINAL ARTICLE

A self-paced brain–computer interface for controlling a robot simulator: an online event labelling paradigm and an extended Kalman filter based algorithm for online training

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Abstract Due to the non-stationarity of EEG signals, online training and adaptation are essential to EEG based brain–computer interface (BCI) systems. Self-paced BCIs offer more natural human–machine interaction than synchronous BCIs, but it is a great challenge to train and adapt a self-paced BCI online because the user's control intention and timing are usually unknown. This paper proposes a novel motor imagery based self-paced BCI paradigm for controlling a simulated robot in a specifically designed environment which is able to provide user's control intention and timing during online experiments, so that online training and adaptation of the motor imagery based self-paced BCI can be effectively investigated. We demonstrate the usefulness of the proposed paradigm with an extended Kalman filter based method to adapt the BCI classifier parameters, with experimental results of online self-paced BCI training with four subjects.

Keywords Brain–computer interface · Self-paced BCI paradigm · Asynchronous · Online training and adaptation · Extended Kalman filter

1 Introduction

A brain–computer interface (BCI) is a communication system in which an individual sends commands to the external world by generating specific patterns in brain signals and a computer detects the brain signal patterns and translates them into commands that accomplish the individual's intention [11, 14]. BCI systems can be categorised into systems operating in cue-paced (synchronous) or self-paced (asynchronous) mode. The majority of the existing EEG-based BCI systems are synchronous [1, 5, 17, 27], in which the analysis and classification of brain signals are locked to predefined time windows. This means that users are supposed to generate commands only during specific periods determined by the BCI system. The advantage of synchronous BCI systems is that the onset of mental activity is known in advance and associated with a specific cue or trigger stimulus, and thus any signal outside the predefined time windows is treated as idling and ignored by the BCI system. On the other hand, self-paced BCI systems offer a more natural mode of human-machine interaction than synchronous BCIs [2, 8, 10, 13, 16, 18, 19, 23]. In self-paced BCIs, no cue is used, the users control the BCI output whenever they want by intentionally performing a specific mental/cognitive task, and the EEG signals have to be analysed and classified continuously. The challenge of self-paced BCIs is that the lack of indications of the user's control intention and timing make it more difficult to design and evaluate a BCI.

In order to make online training or testing possible for a self-paced BCI, current methods for obtaining the user's control intention and timing analyse continuous EEG data consisting of predefined cue-triggered mental states (simulated self-paced BCI) [23], use self-report by subjects [2, 16], rely on subjects to perform a predefined sequence of mental tasks [2, 18], or analyse real movement induced EEG data recorded along with time stamps of actual movements (e.g., from EMG signals) [8, 10, 13]. It should be noted that the above methods are not

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typically suitable for online training purposes, except for the latter approach which assumes that the EEG responses invoked during imagined movements will have enough similarity to those during actual movements which form the 'training' labels. However, realistic BCI applications should ultimately be based on motor imagery or other mental activities rather than real movements. Although it is possible to use self-reporting or rely on subjects to perform a predefined sequence of mental tasks for online training or evaluation, it is limited by the fact that it is difficult for a subject to remember the whole sequence of predefined mental tasks correctly if the testing session is long enough to simulate real-life applications. It is also unsuitable to display the sequence along with the experimental paradigm to assist the user, because it would not be the same experience as using the system at his/her own volition.

This paper proposes a novel motor imagery based self-paced BCI for controlling a simulated robot in a specifically designed environment, which is able to provide the user's control intentions and timing during online experiments, so that online training and adaptation of the motor imagery based self-paced BCI can be effectively investigated. This proposed paradigm for online event labelling works as follows: during online training or testing, subjects are asked to drive a robot to a predefined target position, and the user's control intention at a specific time can be predicted based on the natural way the user would choose his/her path to reach the target. Therefore, it is not required for the user to memorise the predefined sequence explicitly. This paradigm will be described in more detail in the next section.

It is well-known that EEG signals, particularly in EEG-based BCI systems, are non-stationary. The non-stationarities may be caused by the capricious feedback, changes in electrode impedance or dynamically changing environments. To some extent, a realistic BCI system has to be trained online and adaptive even in application phases where the true labels of ongoing EEG trials are unknown. Therefore, adaptation of feature extraction or classification is very important for BCI systems [4, 6, 7, 9, 15, 20–22, 26]. It is a great challenge to train and adapt a self-paced BCI online because the user's control intention and timing are usually unknown to the BCI system. Because the proposed online event labelling paradigm is able to provide user's control intentions during online training, we demonstrate its usefulness in adapting the BCI classifier parameters online, in conjunction with an extended Kalman filter based adaptation method [12].

Online self-paced BCI experiments have been conducted with four subjects, producing promising results on performance improvement by online training in terms of accuracy and deviation of real paths from optimal paths.

2 Methods

2.1 BCI experiments setup

The experiments were carried out with able-bodied subjects who sat on an armchair at 1 m distance in front of a computer screen. The EEG recording was made with a g.tec amplifier (Guger Technologies OEG Austria). Five bipolar EEG channels using 10 electrodes, as shown in Fig. 1, were measured over C3 (FC3 vs. CP3), C1 (FC1 vs. CP1), Cz (FCz vs. CPz), C2 (FC2 vs. CP2), and C4 (FC4 vs. CP4). The EEG was sampled at 250 Hz.

2.2 Synchronous data acquisition and offline training

A simple synchronous BCI paradigm, proposed by the Graz BCI Lab [17], was used to record data for training classifiers offline before online self-paced experiments. The subjects were asked to imagine left versus right hand movements, or be idle (the third class), in each trial. The experiment for each subject consisted of 6 runs with 60 trials each, hence 20 trials for each class in each run. In each trial, the subject relaxed until a green cross appeared on screen at $t = 2$ s (s for second). At $t = 3$ s, a red arrow (cue) pointing either left or right direction appeared on screen for 2 s, or no cue for idle class. The subject's task was to respond to the arrow by imagining left or right hand movements until the green cross disappeared at $t = 8$ s, but the subject was asked to just relax if the trial is for idle class. The order of left and right cues was random, and there was a random interval of 2–3 s between trials.

Logarithmic band power features were extracted from EEG signals and used to classify the imagery movements into left or right class. Frequency bands that give good separation were manually selected for each subject. Using

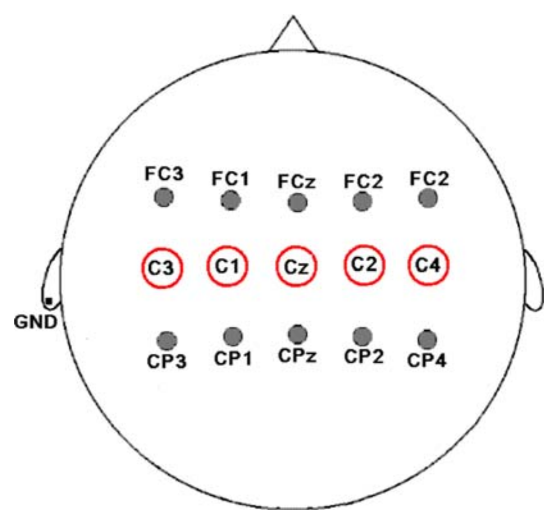

Fig. 1 Five bipolar electrode positions

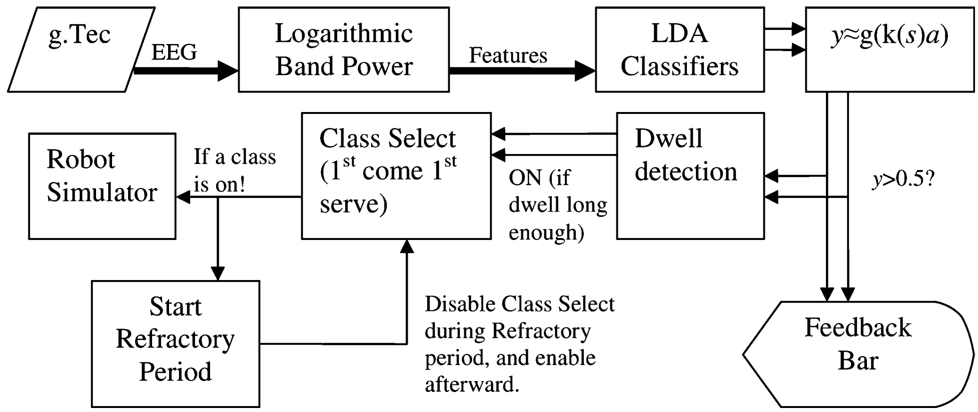

Fig. 2 Online self-paced event detection system

the selected frequency bands, EEG signals were digitally bandpass filtered, squared, averaged over a 1 s sliding window, and a natural logarithm was then applied to obtain the features. Using the extracted features and their corresponding class labels (from the cue signals), two linear discriminant analysis (LDA) classifiers were trained, with one to distinguish left imagery movement from others (right imagery movement or no imagery movement) and the other to separate right imagery movement from others.

It was shown in the BCI competitions 2003 and 2005 that LDA performs as well as (sometimes even outperforms) non-linear classifiers, and almost all the winning classifiers are linear [3]. Therefore, we chose to use and adapt LDA in our BCI design.

2.3 Online self-paced event detection

During online self-paced BCI experiments, the extracted features, which are related to the user's control intention, were continuously classified by the offline trained LDA classifiers and used to control a robot simulator that is described in detail in Sect. 2.4. The online self-paced event detection system used in the experiments is shown in Fig. 2.

Given a feature vector $\mathbf{x}$, the output score (continuous value) of the LDA, with parameter vector $\mathbf{w}$, is calculated as

$$a = \mathbf{w}^{\mathrm{T}}\boldsymbol{\varphi} \tag{1}$$

where

$$\boldsymbol{\varphi} = \begin{bmatrix} \mathbf{x} \\ 1 \end{bmatrix} \tag{2}$$

If the amount of time during which the LDA score, used as a class's confidence value, is above the threshold is longer than the predefined dwell time (>Dwell_Length), a left or right class will be selected as a command to control the simulated robot (otherwise it is the idle class by default), and a refractory period will be switched on at the same time to reject new class/action to be triggered until the refractory period ends. In other words, the "refractory" period is when the triggered command will be ignored, but the count down for dwell detection can start before the end of the refractory period.[1] The threshold and dwell mechanism plus refractory period are implemented here to reduce false positive rate. The refractory period also introduces a competition mechanism for the two LDA classifiers because the selection of one LDA's output will reject the other LDA's output during the refractory period. A simple principle of 'first come, first served' is used here for the class selection (see Fig. 3).

It is important that the thresholds and dwell lengths are chosen for each class and each subject separately in order to enable the system to function with different subjects before online training or testing. This can be achieved by trying various dwell lengths and thresholds when the event detection system with the offline trained classifiers was preliminarily tested with some trial runs before formal online training experiments. We found that it is easier to find a suitable threshold setting if a logistic function is applied to the LDA score, which is defined by

$$y = g(a) = \frac{1}{1 + e^{-\mathbf{w}^{\mathrm{T}}\boldsymbol{\varphi}}} \tag{3}$$

In [24], a method to automatically refine the preset threshold and dwell settings during online training was developed. In this paper, we adopt an alternative approach as we focus on adapting the parameters of the LDA classifiers instead of adapting the threshold and dwell settings.

In our self-paced BCI system y rather than a is used as the classifier output. Alternatively, the classifier output can be produced by thresholding y with a threshold value of 0.5. Because the class labels are from {0, 1}, y is moderated towards 0.5 for better adaptation effect, as suggested in [12]:

$$y \approx g(\mathrm{k}(s)a) \tag{4}$$

[1] According to our experience, all subjects prefer to perform the next mental task before the end of the refractory period, while they see the robot (see Sect. 2.4) is about to reach the next node.

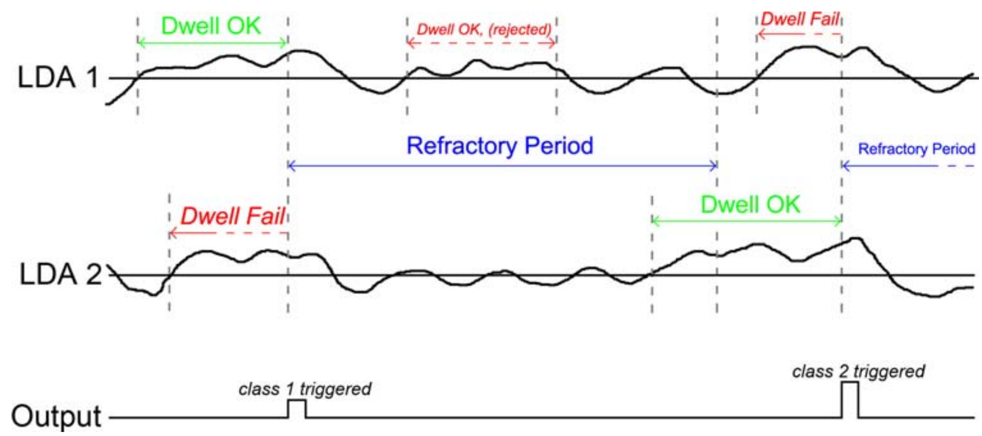

Fig. 3 An example of the "first come, first served" principle for two classes. LDA 1 has shorter Dwell_Length requirement than LDA 2. LDA 1 was first selected because it reached its Dwell_Length requirement sooner than LDA 2. The counting of dwell detection continues even within the refractory period. LDA 2 was then selected as it met its Dwell_Length requirement. It can be seen that the refractory period rejected any command triggered by the classifiers

$$\mathrm{k}(s) = \frac{1}{\sqrt{1 + \pi \frac{s^2}{8}}} \tag{5}$$

$$s^2 = \boldsymbol{\varphi}^{\mathrm{T}} \boldsymbol{P} \boldsymbol{\varphi} \tag{6}$$

where $\boldsymbol{P}$ is the variance of $\mathbf{w}$.

2.4 A robot simulator and its specifically designed environment

During online training, information about the user's control intention and timing is needed so that adaptation algorithm can be executed online. In the case of online testing where no adaptation, takes place, user's control intention is used for system evaluation. For this purpose, a robot simulator, which runs in a specifically designed environment, as shown in Fig. 4, is proposed and implemented for online self-paced BCI experiments.

The environment is filled with hexagon grids, and the robot movements are railed to the grid line. The robot simulator executes two commands: "turn left then move forward to the next node" or "turn right then move forward to the next node", thus the online event detection system operates in 2-class mode. The time required for the robot to move from one node to the next node is the same as the

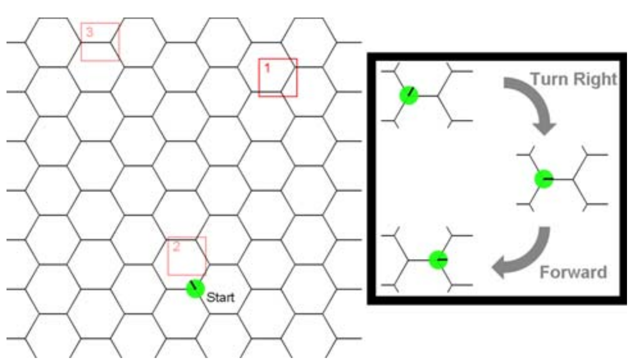

Fig. 4 (left): Hexagon grid, where the circle with a heading indicator represents the robot and the squares are targets. (right): Illustration of a "turn right then move forward to the next node" command executed

refractory period, so the user knows that the refractory period is about to end as the robot is about to reach the next node. The robot simulator stops and waits for the next left/right command after executing a command. The task of the BCI control is to drive the simulated robot to a given target position. There is no obstacle in the environment, but the user is supposed to drive towards the target all the time. An explicit instruction is given to subjects:

Instruction *At each node, the user should always turn the robot to the direction that the target is located.*

In the scenario given in Fig. 4, according to the Instruction, the ideal sequence of commands to control the robot from "start" to the first target should be: right, left, right, left, right, right, left, left, right and right. However, it is unnecessary for the subjects to plan the whole sequence of commands in advance, as long as the subjects follow the Instruction at each node. The simulator executes every detected command, including false positive ones, so it is possible to simulate a real-life scenario where the subjects make mistakes (or the system is at fault) and then try to correct them at the next node to ensure the target would be achieved eventually, maybe via different paths.

By assuming that a subject will follow the Instruction, it is possible to predict what command the subject tries to send or what motor imagery movement the subject tries to perform at each node. This prediction can be used to check whether the event/command detected by the BCI is true positive or false positive.

The key part of the user's intent prediction is the calculation of the target direction at each node. Since the robot position and target position are known, the current direction of the robot, *CD*, and the target direction, *TD*, with respect to the centre of the robot can be calculated by:

$$\begin{aligned} CD &= \operatorname{atan2}(x_2 - x_1, y_2 - y_1) \\ TD &= \operatorname{atan2}(x_{\mathrm{T}} - x_1, y_{\mathrm{T}} - y_1) \end{aligned} \tag{7}$$

where $(x_1,\ y_1)$ is the location of the robot, $(x_2,\ y_2)$ is the robot heading point, and $(x_{\mathrm{T}},\ y_{\mathrm{T}})$ is the target location.

Prediction can be obtained by comparing the values of *CD* and *TD*. However, because the output of atan2 lies in the closed interval $[-\pi, \pi]$, simple comparison is only valid when both *CD* and *TD* are >0 or <0. To avoid this limitation, *CD* and *TD* are rotated together to a point where *CD* is aligned to the axis of $-\pi/\pi$, as shown in Fig. 5. The rotation angle d*A* is as follows:

$$\mathrm{d}A = \begin{cases} \pi - CD & CD > 0 \\ -\pi - CD & CD \leq 0 \end{cases} \tag{8}$$

After rotation the target direction is updated as follows:

$$TD = TD + \mathrm{d}A \tag{9}$$

$$TD = \begin{cases} TD - 2\pi & TD > \pi \\ TD + 2\pi & TD < \pi \\ TD & -\pi \geq TD \leq \pi \end{cases} \tag{10}$$

Finally, the prediction of the command that the user is supposed to produce at each node can be obtained based on the target direction as follows:

$$\text{predict} = \begin{cases} \text{right} & TD > 0 \\ \text{left} & TD \leq 0 \end{cases} \tag{11}$$

2.5 Extended Kalman adaptive linear discriminant analysis classifiers

In a previous study, we compared three online training methods for adapting parameter values of LDA classifiers in cue-based BCI systems [25]. The first adaptation method regenerates new LDA classifiers based on the incrementally updated mean values and variances of the BCI features of different classes. The second and third are Kalman filter (KF) and extended Kalman filter (EKF) based adaptation methods. Kalman filter based methods (both standard and extended) are preferred because they adjust adaptation speed dynamically with respect to the linear response from the current model. In addition, EKF has an advantage over KF because it allows nonlinear state transition and observation models.

An EKF based dynamic logistic regression was proposed by Lowne et al. [12] for BCI experiments with real-movement based self-paced BCI data, and the method is sample-by-sample online adaptation under a sequential Bayesian learning paradigm. The EKF (which allows for non-linear output link functions) is required because classification naturally uses non-linear logistic functions. This paper employs a similar adaptation algorithm but the system adapts with respect to a batch of averaged features instead of sample-by-sample adaptation. In addition, the generalised linear classifiers with Gaussian basis functions used in [12] are replaced by LDA classifiers. Logistic functions are still used because, as explained in Sect. 2.3, using logistic function output make the threshold setting easier in obtaining class outputs (control commands) than using the LDA score calculated by Eq. 1 and is correct from a decision-theoretic standpoint. In order to make online training more smoothly, the LDA classifiers are initialised with offline training so that the learning rate for online training can be set to a smaller value, but still able to cope with the non-stationary changes in EEG signals.

The offline trained LDAs are adapted during online training with predicted labels or true labels using the EKF based method. They are only updated with the data that have been classified with high confidence. This can be achieved by averaging the input features that triggered the dwell detection at each event. Whenever there is a class {left or right} triggered by the dwell detection, an adaptation will take place. Therefore, $\mathbf{x}$ is replaced by $\bar{\mathbf{x}}$ in Eq. 2 to represent the averaged input features.

Based on the algorithm developed in [12], with some modifications, the first phase is the prediction of the mean and variance of the weight vector of the LDA:

$$\mathbf{w}_{t|t-1} = \mathbf{w}_{t-1|t-1} \tag{12}$$

$$\boldsymbol{P}_{t|t-1} = \boldsymbol{P}_{t-1|t-1} + \boldsymbol{Q}_{t-1} \tag{13}$$

where

$$\boldsymbol{Q}_{t-1} = q_{t-1}\mathbf{I} \tag{14}$$

$$q_{t-1} = \max\{I_{t-1}, 0\} \tag{15}$$

$$I_{t-1} = u_{t-1|t} - u_{t-1|t-1} \tag{16}$$

where subscript $t|t$-1 represents time step t before updating and $t|t$ after updating, $\mathbf{I}$ is an identity matrix, I_{t-1} is information gain from the last updating, and u represents the uncertainty of the classifier output y, defined by

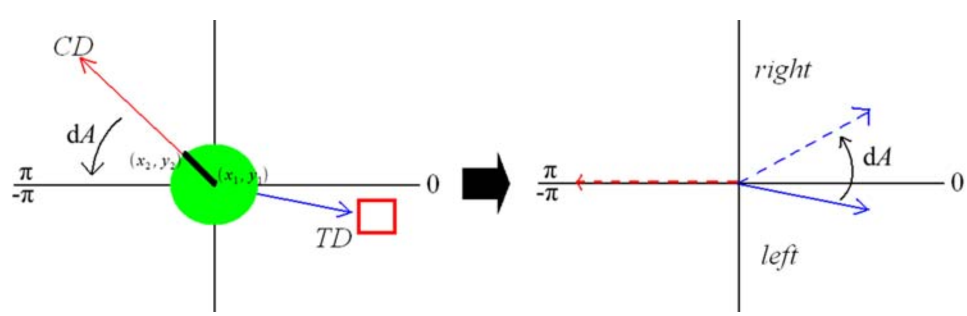

Fig. 5 Rotation of *CD* and *TD* for making a prediction

$$u = y(1 - y) \tag{17}$$

The second phase is the updating of the mean and variance using the Kalman gain as follows [12]:

$$\mathbf{w}_{t|t} = \mathbf{w}_{t|t-1} + \mathbf{K}_t(z_t - y_{t|t-1}) \tag{18}$$

$$\boldsymbol{P}_{t|t} = \boldsymbol{P}_{t|t-1} - \mathbf{K}_t u_{t|t-1} (\boldsymbol{P}_{t|t-1}\boldsymbol{\varphi}_t)^{\mathrm{T}} \tag{19}$$

$$\mathbf{K}_t = \frac{\boldsymbol{P}_{t|t-1}}{\mathrm{C} + u_{t|t-1} s^2_{t|t-1}} \boldsymbol{\varphi}_t \tag{20}$$

$$s^2_{t|t-1} = \boldsymbol{\varphi}_t^{\mathrm{T}} \boldsymbol{P}_{t|t-1} \boldsymbol{\varphi}_t \tag{21}$$

where $\boldsymbol{\varphi}_t$ is the input feature vector as defined in Eq. 2, $\mathbf{K}_t$ is the Kalman gain, C is a learning parameter (Learning rate is smaller with a larger C value), and z_t is the label of the input data at time t, which is the event label prediction as described earlier:

$$z_t = \begin{cases} 1 & \text{true positive detected} \\ 0 & \text{false positive detected} \end{cases} \tag{22}$$

For the next iteration, the mean and variance of the weight vector are set as follows:

$$\mathbf{w}_{t-1|t-1} = \mathbf{w}_{t|t} \tag{23}$$

$$\boldsymbol{P}_{t-1|t-1} = \boldsymbol{P}_{t|t} \tag{24}$$

It should be noticed that if an event is detected, $y_{t|t\text{-}1}$ should be larger than 0.5, and $y_{t|t}$ should be close to z_t. As can be seen from Eq. 18, when a true positive event is detected, the algorithm enhances the classifier with the averaged features that triggered dwell detection. On the other hand, the system makes correction if the detected event is false positive, where $z_t = 0$ while $y_{t|t-1} \geq 0.5$.

3 Results

After offline training, the online self-paced BCI described in the previous section was tested with four subjects: two subjects (S10 and S11) with experience of online synchronous BCI experiments and two subjects (S2 and S3) with no online BCI experience, who participated in our previous study on offline synchronous BCI [25].[2] Four runs of online self-paced BCI experiments were carried out with each subject. In each run, a subject was asked to drive the robot simulator to reach three targets. However, the subject saw only one target at a time. The current target disappeared as soon as the robot reached it, and the next target appeared at the same time. The robot's starting point and target positions were fixed between runs in order to compare performance in a fair manner later with other subjects. The scenarios used in the experiments are shown in Fig. 4. The number of true positive (TP), the number of false positive (FP), and events triggered were recorded. The performance is evaluated by accuracy (ACC) defined as [TP/(TP + FP)], speed defined as the number of events triggered per minute (E/min), and the total number of triggered events (TP + FP) in comparison with the minimum number of events (ME) needed for the robot to reach the targets. Number of adaptations (#A) is the sum of TP and FP. The difference between the total number of events triggered and the minimum number of events needed indicates the deviation of the real path from the optimal path of the robot.

Table 1 Parameter settings for online self-paced BCI experiments

	Frequency bands	Dwell_Lengths (L, R)	Thresholds (L, R)
S2	7–15 Hz and 15–25 Hz	1.4 s, 1.4 s	0.51, 0.50
S3	7–15 Hz and 30–35 Hz	1.6 s, 1.6 s	0.50, 0.50
S10	11–14 Hz and 15–25 Hz	1.6 s, 2.0 s	0.45, 0.48
S11	11–14 Hz and 15–30 Hz	1.4 s, 1.6 s	0.50, 0.50

Prior to formal online experiments, the subjects took some trial runs with the presence of the experiment conductor, so that they understood the paradigm and requirements and a suitable parameter settings could be found. The procedure of trial runs is that the threshold and dwell time for each class were first set at an initial setting,[3] and then these values were adjusted separately[4] for each class by the experiment conductor based on his experience and subject's feedback. Two frequency bands were chosen manually based on offline training data, and band power features were extracted online as described in Sect. 2.2. Therefore, the dimension of the feature vector is 10, since two features were extracted for each channel and all five channels were used. Parameter settings for each subject is summarised in Table 1.

The system was first tested without online training. Each subject performed two runs. The results for performance evaluation are shown in Table 2 (top). The above experiment was repeated, except that the first part of the experiment (from start to the first target) was used for online training. During online training, the system adapts the classifier parameters based on the method described in

[2] In that study, nine subjects participated in an offline BCI experiment. Five were invited for further online experiment, but only three agreed and only two of them were able to control the online self-paced BCI system at a reasonably satisfactory level.

[3] For this experiment, the initial setting for threshold is 0.5, and 1.5 s for dwell time. The first author is the only experiment conductor.

[4] In our experience, subjects usually found themselves easier to control one class over another class, so it is reasonable to assign threshold and dwell time differently.

Table 2 Results from online self-paced BCI experiments: performance without online training (top) versus performance with online training (bottom)

	Start to 1st target					1st to 2nd target					2nd to 3rd target				
	TP	FP	ME	ACC (%)	*E*/min	TP	FP	ME	ACC (%)	*E*/min	TP	FP	ME	ACC (%)	*E*/min
S2—1st run	24	10	10	70.59	5.02	8	0	8	100.0	4.36	14	4	10	77.78	6.21
S2—2nd run	16	8	10	66.67	6.34	20	6	8	76.92	6.24	10	0	10	100.0	4.96
S3—1st run	12	4	10	75.00	6.91	10	3	8	76.92	6.67	14	3	10	82.35	7.44
S3—2nd run	17	5	10	77.27	6.41	20	7	8	74.07	6.51	22	9	11	70.97	6.91
S10—1st run	14	2	10	87.50	8.00	8	1	8	88.89	8.44	10	1	10	90.91	8.36
S10—2nd run	9	1	10	90.00	8.21	12	4	8	75.00	8.07	10	1	10	90.91	8.25
S11—1st run	17	6	10	73.91	6.48	7	1	8	64.71	8.00	21	12	10	63.64	5.59
S11—2nd run	9	2	10	81.82	7.02	10	3	8	76.92	6.19	23	10	11	69.70	6.19
S2—Average=	20	9		68.63	5.68	14	3		88.46	5.30	12	2		88.89	5.59
S3—Average=	15	4.5		76.14	6.66	15	5		75.50	6.59	18	6		76.66	7.18
S10—Average=	12	1.5		88.75	8.11	10	2.5		81.95	8.26	10	1		90.91	8.31
S11—Average=	13	4		77.87	6.75	8.5	2		70.82	7.10	22	11		66.67	5.89

	Start to 1st target (online Training)						1st to 2nd target					2nd to 3rd target				
	TP	FP	ME	ACC (%)	#A	*E*/min	TP	FP	ME	ACC (%)	*E*/min	TP	FP	ME	ACC (%)	*E*/min
S2—3rd run	9	1	10	90.00	10	5.71	15	5	8	75.00	3.03	10	2	10	83.33	3.05
S2—4th run	11	3	10	78.57	14	7.18	18	12	8	60.00	6.02	10	2	10	83.33	5.71
S3—3rd run	9	1	10	90.00	10	5.17	7	1	8	87.50	2.45	9	2	10	81.82	1.37
S3—4th run	10	2	10	83.33	12	6.10	19	6	8	76.00	3.06	14	4	10	77.78	2.60
S10—3rd run	10	0	10	100.0	10	9.04	8	0	8	100.0	8.28	13	2	10	86.67	7.38
S10—4th run	12	2	10	85.71	14	7.43	10	1	8	90.91	7.76	16	4	10	80.00	6.74
S11—3rd run	10	2	10	83.33	12	7.66	7	1	8	87.50	7.50	9	4	10	69.23	6.04
S11—4th run	9	3	10	75.00	12	7.92	14	8	8	63.64	5.92	19	11	10	63.33	6.38
S2—Average=	10	2		84.29	12	6.45	17	8.5		67.50	4.53	10	2		83.33	4.38
S3—Average=	9.5	1.5		86.67	11	5.64	13	3.5		81.75	2.76	12	3		79.80	1.99
S10—Average=	11	1		92.86	12	8.24	9	0.5		95.46	8.02	15	3		83.34	7.06
S11—Average=	9.5	2.5		79.17	12	7.79	11	4.5		75.57	6.71	14	7.5		66.28	6.21

Sect. 2.5. The results with online training are given in Table 2 (bottom).

Note the results from the experienced subjects (S10 and S11) first. S10 made some very good runs when no adaptation was employed, with average accuracy of 88.75, 81.95, and 90.91% in reaching the 1st, 2nd, and 3rd targets, respectively. Nevertheless, while the experiment was carried out with adaptation, this subject performed an almost perfect third run with 100% accuracy when travelling from start to the 1st and 2nd targets. With adaptation, the average accuracies in reaching the three targets are 92.86, 95.46, and 83.34%, respectively. These results have shown noticeable improvement except for the final part of each run when the subject was trying to reach the third target. It is also interesting to notice that this subject made most mistakes at the final part of the third and fourth runs. It might be because of tiredness or possible increase in stress as subjects achieve proximity to the goal. Subject S11 achieved average accuracy of 77.87, 70.82, and 66.67% in reaching the 1st, 2nd, and 3rd targets, respectively, when no adaptation was employed. This subject could control the system reasonably well, but the performance was poor in the final part of the 1st and 2nd runs, with 11 mistakes and 33 events triggered on average in each run while the travel distance could be covered by 10 or 11 events. The system was then switched to with adaptation, and the average accuracy was increased to 79.17 and 75.57% in reaching the 1st and 2nd targets, respectively. The performance is slightly decreased to 66.28% in the final part of the runs, but with less number of triggered events.

The performances of the non-experienced subjects (S2 and S3) are generally poorer than the experienced subjects, in terms of accuracy and speed, when no adaptation was employed. Without adaptation, S2 finished the first half of the experiment with average accuracy of 68.63, 88.46, and 88.89% in reaching the 1st, 2nd, and 3rd targets, respectively, and S3 achieved average accuracy of 76.14, 75.50, and 76.66% in reaching the 3 targets, respectively. For the second half of the experiment, where the first part of each run was operated with online training, S2 achieved average accuracy of 84.29, 67.50, and 83.33% when reaching 1st, 2nd and 3rd targets, respectively, no obvious performance improvement observed, but S3 produced consistent performance improvement, achieving average accuracy of 86.67, 81.75, and 79.80% in reaching the 3 targets, respectively. It is also noticed that in general the non-experienced subjects were slower than the experienced subjects with lower events per minute (*E*/min).

4 Conclusion and discussion

This paper presents a self-paced BCI system with online training for controlling a simulated robot. Taking up the challenge in online training of self-paced BCI systems, this paper has proposed a novel method for providing information about class labels (user's control intention and timing), which is essential for training and adapting self-paced BCIs so as to improve the performance. This paper has also tested the EKF based method for online adaptation of all the parameters of the LDA classifiers in the BCI system. Instead of using sample-by-sample labelling, which is hard to achieve in self-paced BCIs, we adopt event-by-event labelling and a slower learning process. Initial experimental results have shown the effectiveness of the proposed methods. More experiments will be conducted to further justify the methods in self-paced BCI applications, especially the effectiveness of adaptation after long-term usage.

When an intentional control is misclassified by the current model, the system generates a false positive detection or fails to detect at all. Although the proposed paradigm is able to recognise false positive detections and adjust accordingly, the system requires external mechanism to deal with the situation when the system fails to make detection (i.e., False Negative). In addition, an ideal system should also be able to suspend the adaptation process when it is known to the system that (via some sort of detection from the EEG or other physiological signals) the temporary mistakes made by subjects are due to being tired or stressed, or loss of concentration, because adaptation under these conditions would introduce noise to the current model.

The current work is limited to two classes, but the proposed experimental paradigm can be easily extended to multiple classes. Further research will be focused on self-paced BCIs with unsupervised online adaptation when event labels are unavailable.

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